FORMULA 1
RACE INTELLIGENCE DASHBOARD

Project Documentation & Analysis Report

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Institution	Medi-Caps University
Tools Used	Tableau Desktop, Microsoft Excel
Project Type	Data Analytics & Visualization

Prepared for the Formula 1 Race Intelligence Dashboard project.

1. Executive Summary

This project uses Tableau to explore Formula 1 racing data through visualizations and dashboards. The workbook brings together information about race results, drivers, championship standings, circuits, lap intelligence and qualifying results. The goal is to make racing performance easier to explore through comparisons and interactive views.

This report documents the workflow and planned checks. Numerical conclusions should be copied from the final workbook only after validating the filters, relationships and aggregations.

2. Problem Statement

Formula 1 data is spread across categories such as drivers, teams, circuits, race outcomes, lap times and qualifying performance. Reviewing separate tables can make comparisons difficult. This project organizes selected information in Tableau to support visual exploration and analysis.

3. Objectives

• Connect the supplied Formula 1 Excel sheets in Tableau.

• Review data types, field names and relationships between tables.

• Create visualizations for drivers, teams, circuits, race results, lap times and qualifying performance.

• Combine selected worksheets into dashboards.

• Test filters, chart calculations, labels and dashboard usability.

• Document the workflow and communicate findings based on verified results.

4. Dataset & Data Sources

The source Excel workbook used during setup contained these sheets:

Sheet	Purpose
race_results	Race results and driver performance fields.
championship_standings	Season, driver, team, points, wins and standings.
circuits	Circuit identifiers, locations and circuit characteristics.
lap_intelligence	Lap-level information, lap times and tyre-related fields.
qualifying_results	Qualifying performance and positions.

Record the original dataset source, date range, license and row counts from the dataset documentation. Do not publish raw data unless its license permits redistribution.

5. Tools & Technologies

• Tableau Desktop – data modeling, worksheets, calculated measures and dashboards.

• Microsoft Excel – source workbook and sheet organization.

• GitHub – planned version control and project presentation, if required.

6. Data Preparation & Modeling

The Excel workbook was connected to Tableau using the Microsoft Excel connector. Relationships were configured between the logical tables using matching fields available in the source data. Examples used during setup included Circuit Id between race_results and circuits, Race Id between race_results and lap_intelligence / qualifying_results, and Season plus Driver between race_results and championship_standings.

Validation note: Confirm that relationship keys match in both tables and that values are consistent. Relationships can create duplicated or missing totals if keys are not unique or do not match.

• Check field data types (dates, numeric measures, dimensions and identifiers).

• Inspect null values and decide whether to exclude, retain or explain them.

• Check duplicate records and unexpected row multiplication.

• Verify aggregation choices such as SUM, AVG, COUNT and COUNTD.

• Compare selected chart totals with the source data or an independent calculation.

7. Worksheets & Visualizations

No.	Worksheet / analysis topic
1	F1 Driver Championship Points
2	F1 Team Championship Points
3	F1 Racing by Country / Circuits by Country
4	F1 Race Results by Driver
5	F1 Average Lap Time by Driver (not pit-stop time)
6	F1 Average Qualifying Position by Driver
7	F1 Driver Reliability / DNF Analysis
8	F1 Tyre Compound / Tyre Strategy Analysis
9	F1 Weather Impact Analysis
10	F1 Circuit Distance Analysis
11	F1 Race Duration Analysis
12	F1 Safety Car Analysis
13	F1 DRS Overtakes Analysis
14	F1 Fastest Lap Analysis
15	F1 Positions Gained Analysis

Chart accuracy notes: Worksheet 5 was built using average lap time because a confirmed pit-stop-time field was not visible in the fields shown. Do not describe it as pit-stop analysis unless you replace the measure with a verified pit-stop field. Weather comparisons show points by weather category; they do not prove weather caused the differences. Check circuit and race aggregations before interpreting totals.

8. Dashboard Design

The Tableau dashboards combine selected worksheets into a visual workspace. The main dashboard includes driver and team championship comparisons and other race-performance charts. A second dashboard was also created for additional race-performance analysis. The layout was assembled using tiled objects.

• Use clear dashboard and worksheet titles.

• Avoid overcrowding; keep labels readable.

• Use consistent colors and formatting.

• Add useful filters such as Season, Driver, Team or Country where available.

• Test dashboard filter actions and confirm they affect the intended charts.

9. Testing & Quality Checks

• Data connections: all sheets load and relationships use matching keys.

• Chart calculations: measures use appropriate aggregations and no accidental duplicate pills.

• Filters: Season, Driver, Team and Country show valid values and update charts correctly.

• Null handling: nulls are handled intentionally.

• Dashboard usability: charts are readable, aligned and not clipped.

• Saving: packaged workbook opens on another computer with required data.

10. Key Findings

Add three to five findings from the final Tableau workbook. State the season and filters used, and quote exact values only after verifying them.

• Finding 1: _____

• Finding 2: _____

• Finding 3: _____

11. Conclusion

The project demonstrates how Tableau can organize Formula 1 data into visual comparisons across drivers, teams, circuits, race results, qualifying and other performance topics. The dashboards provide a starting point for exploring the dataset. Reliable conclusions depend on correct relationships, appropriate aggregations and final workbook testing.

12. Future Scope

• Add verified pit-stop analysis if pit-stop data is available.

• Improve interactivity with tested filter actions and parameters.

• Add season-over-season comparisons and more detailed driver/team trends.

• Publish a Tableau Public version if the workbook and dataset can be shared publicly.

• Create a short screen-recorded demonstration and include its link if required.

13. Submission Checklist

• Final Tableau packaged workbook (.twbx).

• Final dashboard screenshots.

• This report, updated with student details and verified findings.

• 5–7 minute project explanation video, if required by SkillWallet.

• GitHub repository link and README, if required.

• Tableau Public link, if publishing is permitted and required.

Appendix: Before submitting

Fill in the student details on the cover page, add the dataset source and date range, insert final dashboard screenshots if required, and replace the blank key-findings section with observations verified from your workbook. This is a documentation draft based on the workflow discussed; it does not claim that testing, video recording, GitHub upload or public publishing has already been completed.